

# Experimental Burst-Mode Reception of LDPC Forward-Error-Corrected 256Gbps Coherent PON with 24 dB Dynamic Range using a Static-Gain TIA

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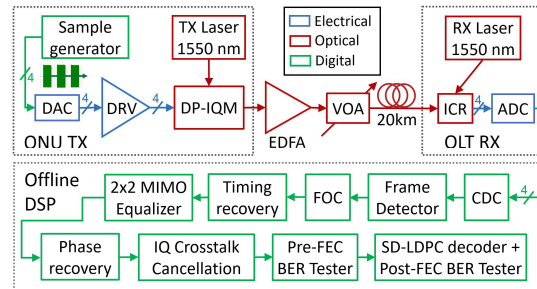
**Abstract** We demonstrate post-FEC error-free upstream burst-mode transmission of 256 Gbps DP-QPSK over 20 km in the C-band. The proposed receiver scheme with commercial ICR and static-gain TIA supports 24 dB of dynamic range (DR) exceeding the typical 20 dB ITU-T requirement for PON systems. ©2026 The Author(s)

## Introduction

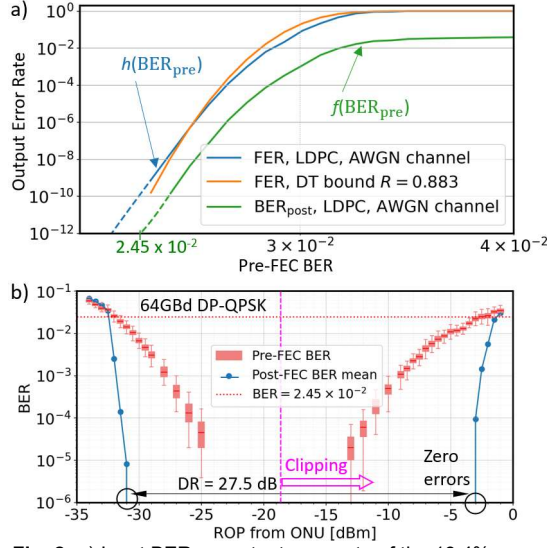
Current research on Very-High-Speed-PON is investigating the enabling technologies for burst-mode coherent PON [1-5]. Adopting coherent technology in burst-mode PON is challenging; commercial coherent receivers (RX), designed for continuous-mode operation, lack burst-mode transimpedance amplifiers (TIA) capable of handling the wide dynamic range (DR) of input burst powers. A coherent burst-mode RX operating at 256 Gbps in the C-band with static-gain TIA and 20 dB DR was demonstrated in [6], assuming error-free operation at a pre-FEC BER below  $2 \times 10^{-2}$ . This high DR is supported by working with clipped signals in the high-power regime, using the clipping-tolerant DP-QPSK modulation scheme. Recent high-speed PON generations [7] use hard-decision or soft-decision (SD) low-density-parity-check (LDPC) code for forward-error-correction (FEC). These codes are designed assuming additive white Gaussian noise (AWGN) channels. This condition is typically well-fulfilled in the low-input-power regime of coherent RXs with adaptive equalizers in continuous mode. However, the signal quality in the high-power regime of a static-gain coherent RX is mainly limited by signal distortions due to TIA nonlinearities and analogue-to-digital converter (ADC) clipping. Similarly, burst-mode operation might introduce non-AWGN channel behaviour in the low-power regime. Hence, it should be investigated whether these signal distortions lead to a significant net coding gain reduction (NCGR) when applying SD-LDPC FEC. In this paper, we measure post-FEC BER at different received optical powers (ROP) using 18.4%-overhead SD-LDPC FEC. The number of measured bits allows for the reliable measurement of post-FEC BER values down to just  $\sim 1 \times 10^{-5}$ . To estimate the post-FEC BER below that value, we propose two methods that leverage empirical distributions of error counts and average information density per codeword, respectively. We show error-free operation within a 24 dB ROP range and derive the NCGR in the low-power and high-power regimes.

## Test Setup

In our previous work [6], we have demonstrated alternating reception of soft- and loud-bursts using a coherent RX. Since post-FEC BER analysis requires extensive statistics, we simplified the setup to receive bursts from a single optical network unit (ONU). The experimental setup is depicted in Fig. 1. Repetitive bursts at 64 GbD DP-QPSK with 602 ns preamble and 649 ns payload are generated digitally with a 120 GSps digital-to-analogue converter (DAC). The ONU transmitter (TX) comprises a 4-channel driver amplifier (DRV), a dual-polarization IQ-modulator and a 100 kHz laser operating at 1550 nm. The link consists of a variable optical attenuator (VOA) and a 20 km fibre span. The VOA emulates splitting loss and controls the input power into the optical line terminal (OLT) RX. An Erbium-doped fibre amplifier (EDFA) is used to boost the burst power into the VOA to +8 dBm, enabling reception of high-power bursts. The OLT RX consists of a commercial class-40 integrated coherent receiver (ICR) module, a 100 kHz C-band local oscillator, and a 96 GSps ADC. The frequency offset between ONU TX and OLT RX is set to 0.6 GHz. The received waveforms are processed by digital signal processing (DSP) including chromatic dispersion compensation (CDC), frame detection and alignment, frequency offset estimation and compensation



**Fig. 1:** Experimental burst-mode setup and offline DSP for OLT reception range characterization after 20 km of fiber transmission in C-band at 64 GbD DP-QPSK.



**Fig. 2:** a) Input BER vs. output error rate of the 18.4%-overhead SD-LDPC FEC for an AWGN channel. By comparing to the DT bound, we find that implementation loss degrades the code performance to an effective code rate of  $\sim 0.883$ . b) Pre-FEC and post-FEC BER vs. ROP. A box plot shows the pre-FEC BER distribution of the bursts for each ROP.

(FOC), timing recovery, constant-amplitude-zero-autocorrelation-based 2x2 MIMO equalization, phase recovery, IQ crosstalk cancellation, and SD-LDPC decoder. The BER is measured before FEC decoding (pre-FEC) and after FEC decoding (post-FEC). The simulated BER transfer function  $f$ , describing the relationship between the input BER at the SD-LDPC decoder (implemented in floating-point arithmetic) and the resulting output BER is depicted in Fig. 2(a) for an AWGN channel. The average pre-FEC BER threshold, for achieving a post-FEC BER of  $1 \times 10^{-12}$ , is found to be  $2.45 \times 10^{-2}$  through extrapolation.

#### Direct Post-FEC BER Measurements

To measure the supported DR of the OLT RX, we sweep the ROP. Fig. 2(b) depicts the measured pre-FEC BER and the corresponding average post-FEC BER. For each ROP setting,  $\sim 1150$  bursts are recorded. Their pre-FEC BER distribution is illustrated by a box plot. After FEC decoding of the bursts, we find a post-FEC error-free DR of 27.5 dB. However, due to the offline acquisition, the sample size is insufficient for claiming post-FEC BER operation below  $1 \times 10^{-12}$ . Next, we propose two methods for estimating the post-FEC BER from the measured distributions of the error counts per codeword and average information density per codeword, respectively.

#### Post-FEC BER Estimation Method I via Codeword Error Histograms

This method consists of three steps and is illustrated in Fig. 3(a).

**1) Determining the codeword error distribution:** The number of pre-FEC errors  $n_{\text{err}}$  within each

codeword is determined, forming a histogram.

**2) Probability density function (PDF) fitting:** PDF<sub>1</sub> is fitted to the histogram. For an AWGN channel, the PDF is approximated by a Gaussian.

**3) BER estimation:** Integrating the fitted PDF<sub>1</sub> for  $n_{\text{err}} > 423$  (corresponding to  $\sim 2.45 \times 10^{-2}$  pre-FEC BER) would yield the frame error rate (FER). To obtain the post-FEC BER, we weigh the PDF with  $f(\text{BER}_{\text{post}})$  from Fig. 2(a) and integrate over the codeword length.

$$\text{BER}_{\text{post}} \approx \int_0^n \text{PDF}_1(n_{\text{err}}) \cdot f\left(\frac{n_{\text{err}}}{n}\right) dn_{\text{err}} \quad (2)$$

The integral is illustrated as red area in Fig. 3(a).

#### Post-FEC BER Estimation Method II via Average Information-Density Histograms

This method consists of four steps and is illustrated in Fig. 3(d).

**1) Determining the Effective Code Rate:**

Practical decoders exhibit an implementation penalty  $\Delta$ . To account for it, we introduce an effective code rate  $R_{\text{eff}}$ . We compare the simulated FER of the LDPC code to a theoretical benchmark, the dependency testing (DT) bound [8]:

$$\text{FER}_{\text{DT}} \approx \mathbb{E}[g(\bar{i})] = \mathbb{E}[2^{-n(\bar{i}-R)^+}], \quad (1)$$

where  $n = 17280$  is the length of the codeword and  $\bar{i}$  is the average information density of a codeword, see Eq. (3), with  $(x)^+ = \max(x, 0)$ . Specifically, the rate  $R$  in (1) is adjusted such that the DT bound matches the simulated FER of the LDPC code. This yields,  $R_{\text{eff}} = R + \Delta \approx 0.883$  as illustrated in Fig. 2(a).

**2) Average Information Density Distribution:**

Assuming a memoryless binary-input AWGN channel, the single-letter information density  $i(x, y)$  describes the amount of information obtained from one channel use and is a random variable. It is computed as [9]:

$$i(x, y) = 1 - \log_2(1 + e^{-x \cdot \text{LLR}(y)}), \quad (2)$$

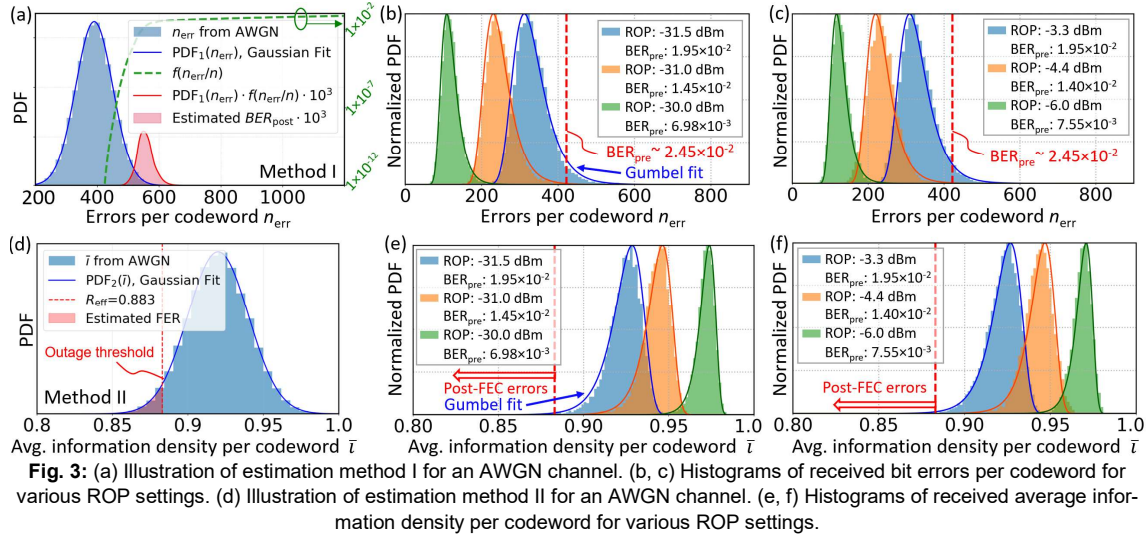
where  $x$  and  $y$  are the transmitted one-dimensional symbol and the corresponding received value, respectively. The log-likelihood ratio (LLR) is computed per DP-QPSK dimension (XI, XQ, YI, YQ) as described in [9]. We define the average information density  $\bar{i}$  of a received codeword of length  $n$  as:

$$\bar{i} = \frac{1}{n} \sum_{l=1}^n i(x_l, y_l) \quad (3)$$

By measuring  $\bar{i}$  of a series of received codewords, a distribution of the channel's "quality" per frame can be created, see Fig. 3(d).

**3) PDF Fitting:** For sufficiently long codewords and AWGN channels, the PDF of  $\bar{i}$  follows a Gaussian shape (Central Limit Theorem [10]).

**4) BER estimation:** For large  $n$ ,  $2^{-n(\bar{i}-R)^+}$  in



Eq. (1) can be approximated by a step function:

$$2^{-n(\bar{i}-R)^+} = \begin{cases} 1, & \text{for } \bar{i} < R \\ 0, & \text{for } \bar{i} > R \end{cases} \quad (4)$$

i.e., the FER is dominated by the event when the average information density  $\bar{i}$  falls below the code rate  $R$  (outage principle) [11]. Hence, with  $R = R_{\text{eff}}$ , the FER is the red area in Fig. 3(d):

$$\text{FER} \approx \int_0^{R_{\text{eff}}} \text{PDF}_2(\bar{i}) d\bar{i} = \Pr(\bar{i} < R_{\text{eff}}), \quad (5)$$

We convert FER to BER using the functions  $f$  and  $h$  from Fig. 2(a).

$$\text{BER}_{\text{post}} = f[h^{-1}(\text{FER})] \quad (6)$$

### Post-FEC Performance Results

We apply methods I and II to the measured data from Fig. 2(b). For every ROP setting, we have a total of  $\sim 10\text{k}$  codewords. The measured histograms of  $n_{\text{err}}$  and  $\bar{i}$  per codeword are depicted for exemplary ROP settings in Fig. 3(b, e) for the low-power regime and in Fig. 3(c, f) for the high-power regime, respectively. We notice that the lab channel does not behave as AWGN since a Gaussian fit is unsuitable. Instead, we find that the long-tailed Gumbel distribution [12] fits our histograms reasonably well. The measured pre-FEC and post-FEC BER and the estimated post-FEC BER are depicted in Fig. 4. Both estimation methods are in good agreement with the measured post-FEC BER in the high-BER

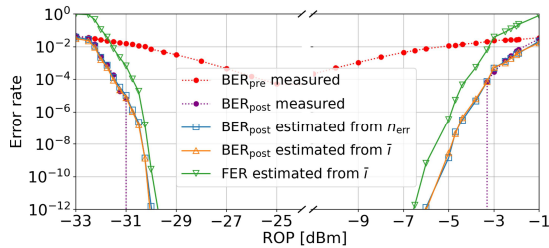
regimes ( $> 1 \times 10^{-5}$ ), where statistics are sufficient. Methods I and II further estimate the post-FEC BER for ROP settings where the sample size is insufficient. Both methods are in excellent agreement and show that the post-FEC BER is below  $1 \times 10^{-12}$  for ROPs  $\geq -30.0$  dBm and  $\leq -6.0$  dBm. Hence, the supported post-FEC error-free DR of the current SD-LDPC burst-mode RX is 24 dB.

We compare the estimated ROP thresholds to the ROP values that have an average pre-FEC BER of  $2.45 \times 10^{-2}$ , i.e., the simulated threshold of the AWGN channel of Fig. 2. We find ROP penalties of 2.0 dB and 3.6 dB for the low-power and high-power thresholds, respectively.

The NCGR is computed as the SNR difference (in dB) between two AWGN channels: Channel 1 has an estimated post-FEC BER of  $1 \times 10^{-12}$ , while channel 2 has the same average pre-FEC BER as our lab channel at the estimated error-free ROP (i.e.  $\text{BER}_{\text{post}} = 1 \times 10^{-12}$ ). We apply this method to the low-power and high-power regimes and find nearly identical NCGR of  $\sim 1.86$  dB and  $\sim 1.76$  dB, respectively. In the low-power regime, the penalty is due to our current burst-mode DSP implementation. Neither the NCGR nor the shape of the histograms in Fig. 3 indicates a dominant impact of clipping. Reducing the DSP penalty is the subject of ongoing research and may reveal details about clipping-related NCGR in the future.

### Conclusions

We demonstrate 20 km upstream burst-mode transmission of 256 Gbps DP-QPSK in the C-band. Our commercial ICR with static TIA gain supports 24 dB DR with post-FEC error-free operation using SD-LDPC coding. We propose two methods for estimating post-FEC BER based on error and information density statistics, both in good agreement with directly measured values. Comparable NCGR is observed in both the thermal-noise-limited region and in the ADC clipping-limited region.



**Fig. 4:** Post-FEC BER vs. received burst power. Values above  $1 \times 10^{-5}$  could be directly measured and agree well with the estimated values from method I and II. We find a post-FEC error free ROP range of 24 dB.

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